A WED-BASED HUMAN RESOURCES EMPLOYEE MANAGEMENT SYSTEM USING FINGERPRINT SCANNER FOR MUNICIPALITY OF MEYCAUAYAN CITY

A Capstone Project Presented to the Faculty of ACLC College of Meycauayan

Meycauayan City, Bulacan

In Partial Fulfillment of the Requirements for the Degree in

Bachelor of Science in Information Technology

IT Capstone 1

UGRD-IT6396

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October 2025

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ACKNOWLEDGEMENT

We would like to express our deepest gratitude to **Mr. Apollos Dailo** and **Engr. Natanael Mendoza** for their continuous guidance, patience, and encouragement throughout the completion of our capstone project. Their valuable advice, constructive feedback, and technical expertise have greatly contributed to the improvement of our proposed system and documentation.

We would also like to extend our heartfelt thanks to our panelists, **Prof. Apollos Dailo, Prof. Justine Santos,** and our **School Consultant, Mrs. Josefina C. Blanco**, for generously sharing their time, knowledge, and helpful insights. Their guidance and support have been instrumental in enhancing the quality and presentation of our work.

Our appreciation also goes to our school, especially **Ms. Jean Carmel Mendoza,** our School Director, and **Mr. Meljohn, Navales** our Proofreader. We are truly thankful for their support and assistance throughout the entire process of our research and documentation. Their understanding, encouragement, and motivation inspired us to remain determined and focused on achieving our goals.

We are sincerely grateful to our **classmates** and **friends** who have shared their ideas and offered moral support throughout the development of this study. Their cooperation and encouragement helped us overcome the challenges we encountered along the way.

Finally, we would like to thank **God** Almighty for the wisdom, strength, and perseverance He has given us throughout this journey. Without His divine guidance and blessings, this project would not have been possibl

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List of Acronyms

ATmega328P - Microcontroller used in Arduino Uno
BSIT - Bachelor of Science in Information Technology
C++ Programming Language (C Plus Plus)
CBA - Cost-Benefit Analysis
CSS - Cascading Style Sheets
DC - Direct Current
DFD - Data Flow Diagram
DPI - Dots Per Inch
GB - Gigabyte
HR - Human Resources
HR Admin - Human Resource Administrator
HRIS - Human Resource Information System
HRM - Human Resource Management
HTML - HyperText Markup Language
IDE - Integrated Development Environment
IoT - Internet of Things
IT - Information Technology
JS - JavaScript
JST - Japan Solderless Terminal (type of connector wire)
LAN - Local Area Network
NVMe - Non Volatile Memory Express
PC - Personal Computer
PDS - Personal Data Sheet
PP - Payback Period
R307 - Model of Fingerprint Scanner (Optical Type)
RAM - Random Access Memory
RDBMS - Relational Database Management System
ROI - Return on Investment
SDLC - System Development Life Cycle
SSD - Solid State Drive
TAM - Technology Acceptance Model
TB - Terabyte
UART - Universal Asynchronous Receiver Transmitter
UI - User Interface
UX - User Experience
VS - Code Visual Studio Code
VRAM - Video Random Access Memory

CHAPTER I

INTRODUCTION

1.1 Background of the Study

The Municipality of Meycauayan City serves a diverse and growing population through its several departments and barangays, which manage public services, keep records, and implement community projects. Despite its commitment to providing services to its residents, the city continues to rely heavily on traditional methods such as physical logbooks, manual record-keeping, and the use of third-party software to maintain and manage essential documents. These characteristics have led to file format incompatibilities, missing data, unstandardized naming conventions, and workflow delays.

Currently, the continued use of manual and fragmented systems has caused difficulties in accurately verifying records, retrieving past data, and generating timely reports. Information is often duplicated, scattered across different formats, or stored in systems with no unified structure. Employees are burdened with repetitive tasks, multiple system entries, and cross-checking procedures, which affect productivity and efficiency. These problems are most apparent during audits, compliance checks, or when urgent decisions are needed

wherein outdated or misplaced records can lead to poor decisions or service delays.

Recognizing the need for digital transformation, this study proposes the development of a **Web-Based Human Resources Employee Management System using a Fingerprint Scanner** for the Municipality of Meycauayan City. This system will aim to unify the physical logbook, third-party applications, and digital storage platforms into one streamlined solution. It will feature automation of data entry, integration of staff and systems, standardized labeling and folder structures, and a centralized access point for verification and reporting. One of the system's key features is the use of a fingerprint scanner to monitor employee attendance, ensuring precise, real-time recording of time-in and time-out while avoiding timekeeping fraud such as proxy logging or "buddy punching."

By introducing this system, the city government can expect to reduce manual workloads, eliminate redundancy, enhance data integrity, and improve the flow of information across departments. Reports and updates can be delivered more accurately and efficiently, while decision-making processes will benefit from organized and real-time access to reliable data. In the long run, this technological solution will contribute to a more responsive, accountable, and efficient local governance model, better equipped to meet the growing demands of the Meycauayan community.

1.2 Statement of the Problem

Based on the data analysis and survey we gathered from the Human Resources Department in Municipality of Meycauayan, the problems are the following:

1. The current system of physical logbooks and third-party applications is filled concurrently without a definitive log, causing manual work and potential mistakes. The information is captured in different formats, leading to missing details and spelling errors. This can complicate confirming the correct record version and potentially lead to incorrect decisions based on estimates and assumptions.
2. The current attendance system in the Municipality of Meycauayan City is prone to fraud and inaccuracy due to the lack of secure identity verification. This leads to unreliable records and reduced accountability. A fingerprint-based system is needed to ensure accurate, real-time monitoring and improve transparency in employee attendance.
3. Employees input data into multiple systems, increasing workload and time waste. This process restricts workflow, ensures data integrity, and can cause productivity issues, confusion, and delays in operations. Companies'

ability to respond to requests is limited due to slow processes, status updates, and report delivery.

4. When past records are saved without a naming convention or folder structure, it becomes very hectic to locate information. Confusing this situation further, records may be physically stored in logbooks and die in the form of spreadsheets or Google Drive documents there are no standardized naming conventions for such records, and staff will spend copious time sifting through numerous files, folders, or logbook pages just to locate a single entry. All these would lead to delays to the daily workflow and further increase the chances of important relevant data being discarded or retrieved from the outdated version. These situations become even more inefficient during the times of audits, compliance checks, or instant decisions-for example, making a decision about the ongoing project or approval to spend.

These are the points why we proposed **A Web-Based Human Resources Employee Management System using Fingerprint Scanner for the Municipality of Meycauayan City** for the way of solutions to the problem

1.3 Objective of the Study

The objective of this study is to give a solution and lessen the problem of the Human Resources Department in Municipality of Meycauayan City to:

1. Offering a unique workflow that will help merge the physical logbook, some application, and digital storage into one platform. This integration does the elimination of duplicate entries, information errors and ensures integrity so that all access and updating of information are available for verification in support of effective decision-making.
2. To develop and implement a fingerprint-based attendance monitoring system for the Municipality of Meycauayan City that ensures secure identity verification, provides accurate and real-time monitoring of employee attendance, and enhances transparency and accountability in workforce management.
3. The physical and digital records should be standardized for labeling and systematization that includes a clear file naming convention, logical folder ranking, and proper categorizations. This would allow for easy retrieval and decrease the likelihood of using outdated or incorrect information. Records should therefore be arranged for easy search ability when day-to-day operation, audit, or time-impacting decision-making is in place.
4. The establishment of a single, organized system for reviewing and cross-checking records from the physical logbook, an application of third parties, and from digital storage would ensure that all data remain accurate against any inconsistencies or out of date records, while accountability for verification would be clearly assigned roles. This would bring early detection and correction of difference in order to avoid having an impact of the overall reporting, compliance, and critical decision-making

1.3 Conceptual Framework

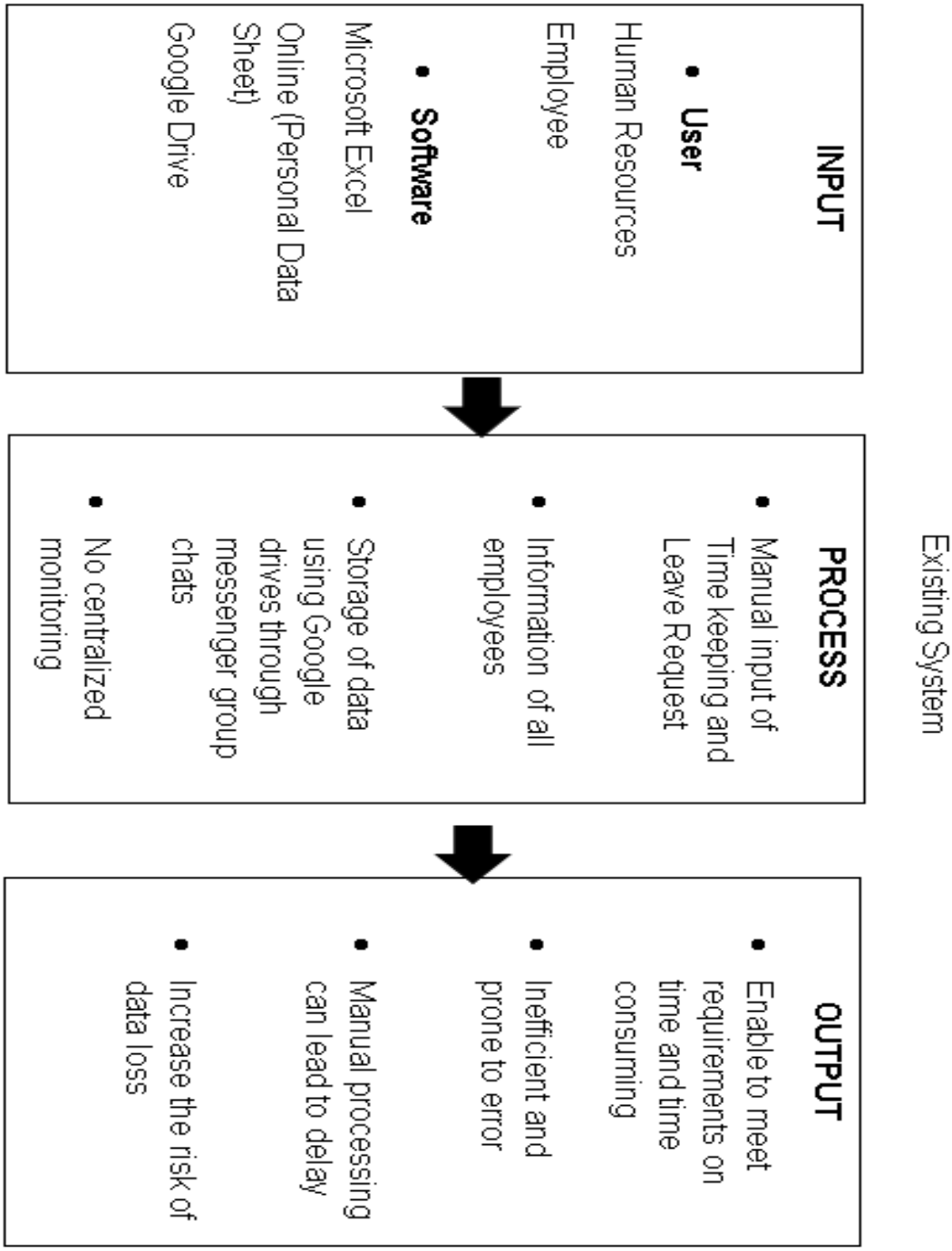


Figure 1.2 – Conceptual Framework of the Existing System

The Existing System shows the current operations of the HR department. Heavily reliance's on manual input of data, documentation and timekeeping using physical logbooks and third party software can lead to inefficient and inaccuracy way of operation.

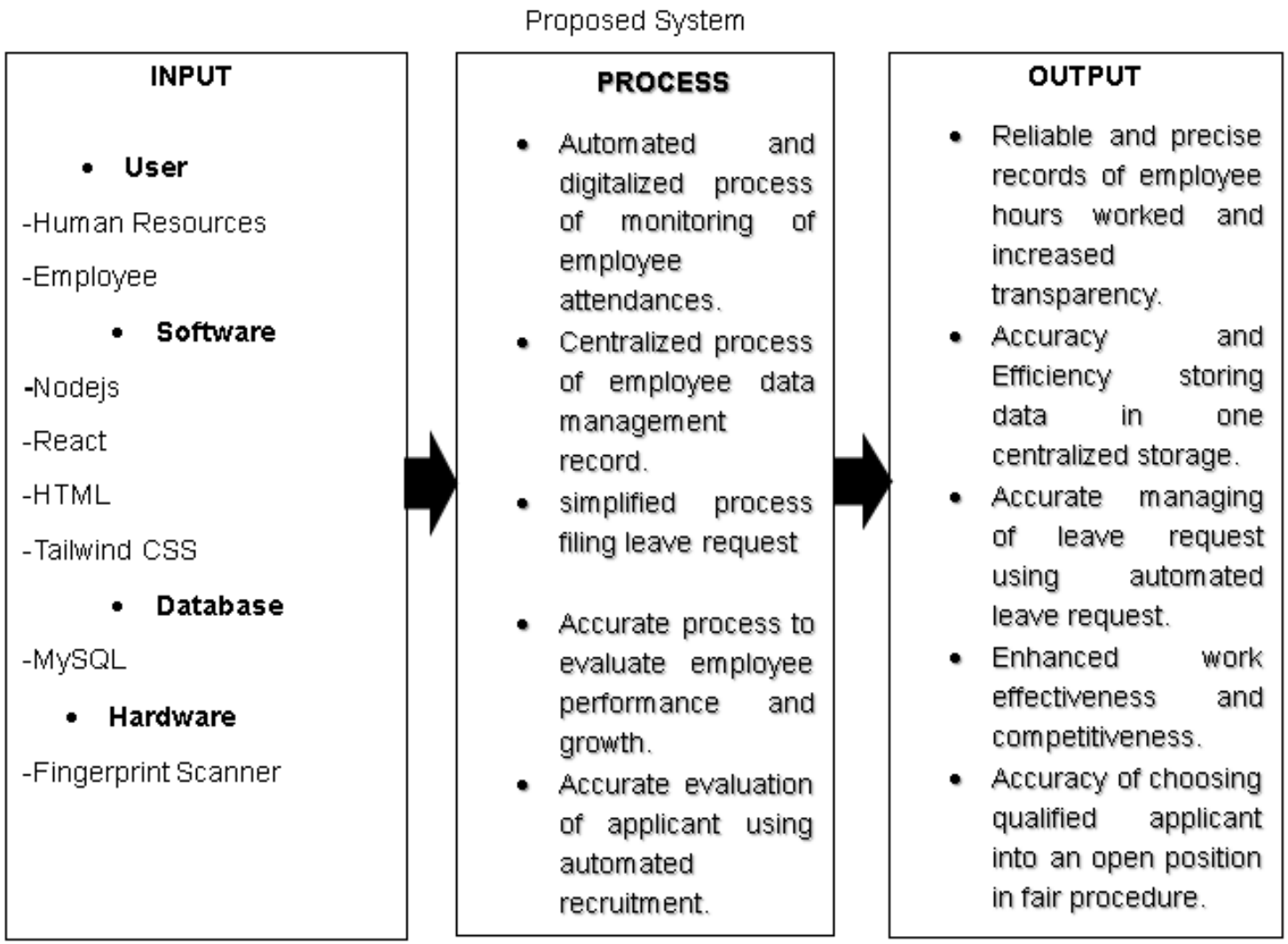


Figure 1.3 – Conceptual Framework of the Proposed System

The proposed system, shows the human resources capability to less the process of relying in the manual process and third-party software, although the system lessen to the third-party software applications, will continue to use in enhances way for exporting and importing file. The system features an automated system that provide more accurate and efficient way of operation for the Municipality of Meycauayan. The system includes, timekeeping to process and monitor employee attendances using fingerprint scanner and result to better worked and attendances transparency. Employee Profiling provides accuracy of employee data in one centralized database. The process of applying for leave requests has been simplified using an automated method. The employee performances evaluation provides an accurate and efficient evaluation of every employee for better effectiveness and competitiveness. The recruitment process ensures accuracy and provides assurances for selecting the most qualified applicant for an open position, adhering to fair hiring procedures.

1.5 Significance of the Study

The **A Web-Based Human Resources Employee Management System using Fingerprint Scanner for the Municipality of Meycauayan City** for **Municipality of Meycauayan** will deliver major significance to stakeholders with

varying objectives and responsibilities. The system is expected to bring substantial improvements in terms of administrative efficiency, data accuracy, and service delivery.

For **Human Resources administrators**, the automation of employee registration, attendance, leaves will optimize daily Human Resources functions. This will minimize manual paperwork, improve accuracy, and enable faster retrieval of employee data for reporting and informed decision-making.

For **employees,** automated systems like for ton leave applications, evaluating employee performance, and improves the accuracy and speed with which these functions are executed. Efficiency and greater levels of transparency will be attained with regard to the handling of personal records and entitlements.

For **management,** the system will automate initial candidate screening as well as interview scheduling and on boarding. This will optimize time, improve candidate monitoring, and enhance the recruitment process.

For **applicants and new hires,** the system will conveniently enable candidate application submission, hiring status tracking, and integration into the company, thus fostering a smooth hiring and on boarding process. This will serve to improve the overall candidate experience and shorten the hiring cycle.

For **company,** the proposed system for producing transformational changes in the organization is replacing scattered, inefficient manual Human Resources process with an automated centralized platform. This brings in overall efficiency, lower human errors, and better data security, thereby further improving the performance and service by the organization.

For **future researcher,** it would provide them exposure to applying their technical and analytical skills towards developing an applicable system to address real problems in an organization. It would give the students practical experience in system analysis, design, development, and implementation.

The web-based system will considerably benefit from a central employee and program management system, which will streamline Human Resources activities, employee records, and recruitment. The system will reduce manual workloads, improve data accuracy, and streamline service delivery. It will also enhance the recruitment process through automated applicant tracking and on boarding. The system will lead to a more efficient work environment, stronger data security, and improved organizational performance.

1.5 Scope and delimitation

Scope

This study aims to create a web-based Employee and Program Management System for Meycauayan City, streamlining Human Resources functions like employee registration, attendance tracking, leave management, performance evaluation, and recruitment, thereby improving administrative efficiency, reducing manual workload, and enhancing data accuracy.

1. **Employee Profiling:** Centralized database for employee records such as personal details, employment history, and status.

2. **Time keeping and Attendance:** Automated attendance monitoring using a fingerprint scanner to ensure accurate tracking and transparency of employee work hours.

3. **Leave Management:** Allows Human Resources staff to manage employee registration, monitor attendance logs, process leave requests. It also enables performance evaluation and creates data-driven reports for managerial use**.**

4. **Recruitment and Applicant Tracking:** Digital submission of applications, candidate monitoring, and streamlined onboarding process for new hires.

5. **Performance Evaluation:** Automated evaluation of employee performance to promote fairness, accuracy, and efficiency in employee assessments.

6. **Reporting and Monitoring:** Generation of real-time reports regarding attendance, leave balances, performance results, and recruitment status for easier decision-making.

Delimitation

While the system enhances Human Resources operations and recruitment within the Municipality of Meycauayan City, it is important to clarify the limitations of the system in terms of scope and functionality:

1. **Department Coverage:** The system will be developed specifically for the Human Resources Department of the Municipality of Meycauayan. Other departments may benefit indirectly but are not within the direct implementation scope of this study.

2. **Biometric Limitation:** The attendance feature will rely solely on the fingerprint scanner; other biometric technologies (e.g., facial recognition, RFID cards) will not be included.

3. **System Access:** The system is designed for use within the city government offices. Remote access outside the municipal premises will depend on internet connectivity and security protocols implemented by the IT department.

4. **Third-Party Integration:** Although the system minimizes reliance on third-party applications, it will still support basic import and export functions (e.g., spreadsheets, reports) but will not guarantee full integration with all external software.

5. **Budget and Infrastructure:** The study will not cover the financial cost of large-scale hardware procurement (e.g., multiple scanners, servers) or long-term system maintenance, which will be the responsibility of the Municipality.

6. **User Training:** The system design will include documentation and user guides, but extensive staff training and change management will not be part of this project.

Definition of Terms

1. **Web-Based System** - A software program that can be accessed via an intranet or the internet via a web browser. The HR management platform

that authorized users in the Municipality of Meycauayan can access is referred to in this study.

2. **Human Resources(HR) -** a division in charge of overseeing services linked to employees, including hiring, compliance, leave processing, attendance, and performance reviews. The digitization of these HR tasks is the main goal of this project.
3. **Employee Management System** -A digital system that helps in managing employee information, including personal records, attendance, leaves, and performance. It streamlines HR tasks and reduces paperwork.
4. **Fingerprint Scanner-** A biometric device that reads an individual's unique fingerprint pattern for identification. In the proposed system, it is used to monitor employee attendance and eliminate proxy logging.
5. **Biometric Authentication -** A method of verifying an individual's identity using biological traits like fingerprints. This ensures secure and accurate attendance tracking in the proposed system.
6. **Proxy Logging (Buddy Punching) -** An act where one employee logs attendance on behalf of another. This fraudulent activity is addressed by implementing fingerprint-based timekeeping.
7. **Attendance Monitoring -** The process of recording the time employees arrive and leave the workplace. In this study, it is automated through the fingerprint scanner to enhance accuracy and transparency.

8. **Employee Profiling -** The creation and management of detailed digital records for each employee, including personal information, employment status, and work history.
9. **Leave Management** - A system that allows employees to request leaves and enables HR to approve, track, and record those requests efficiently.
10. **Performance Evaluation** - A structured process used to assess employee performance over a specific period. The proposed system automates and standardizes this process for fairness and accuracy.
11. **Recruitment and Applicant Tracking** - A digital process for receiving job applications, tracking applicants, and managing onboarding. This function helps streamline the hiring process.
12. **Data Integrity** - Refers to the accuracy and consistency of stored data. The system aims to maintain high data integrity by minimizing manual entry errors and using standardized formats.
13. **Centralized Database -** A unified data storage system where all employee information and HR records are stored, making data retrieval faster and more organized.
14. **Third-Party Software** - External applications not developed by the organization but used to perform certain HR functions. The proposed system minimizes the dependency on such software by consolidating functions internally.

15. **Audit and Compliance** - Processes to ensure that operations and records meet legal and organizational standards. The proposed system supports easier audits through organized and real-time data access.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter highlights the significance of examine relevant studies and literature that reinforces the value of the research. In these studies, we assess the relevant research and literature to gain a solid foundation and comprehension of the study's importance. The researchers will offer a synthesis that takes into account the innovations to enhance comprehension and elucidate the purpose of the study.

2.1 Software Literature and Studies

One the objective of creating the human resources management system is to automate and digitalized the process and the accuracy and efficiency of time without too much reliance on manual in and third-party software. The automated and digitalized human resource employee management system is widely implemented to cater all the needs and in any other types of application platforms including the human resources management department of municipality of meycauayan city. The system will start on creating user interfaces and prototyping phase to plan the design of the systems, before starting to developing phase for faster development.

Timekeeping

The Human Resources is one of the departments inside the municipality of meycauayan city that implemented the employee management system that wanted to be upgrade the current operations of the department. In order lessen the reliance on manual and third-party software applications. The implementation of timekeeping is highly recommended to ensure the accuracy in monitoring the attendances status of the employee. This approach is will be highly recommended to help the day-to-day operation in many different website platform and businesses. According to **Garg V. et al (2018).** Discuss the transformation in technology based-based biometric attendances systems. This study emphasizes the integration human resources management (HRM) in timekeeping of the employee. This study highlights how the biometrics, including fingerprint scanners, enhances the accuracy, reduces proxy attendances, and streamline employee monitoring compared to traditional like manual register, RFID or even third-party software.

Employee Profiling and Centralized Database

The implementation and integration of the employee profiling and centralized data in the business and government systems ensures how the large data stored and properly handled by this study. The record details of every employee in the human resources department are also placed in a third-party

software based on findings from the interview; they are in the computer but have no specific storage for every record. The result of this process should be vital for the integrity of employee information and can lead to data loss. According to **Kavanagh M. et al (2017).** The study of comprehensive edited volume comprising multiple chapters by various author, exploring the theory, design, implementation and future trends of Human Resources Information system (HRIS). The study is detail how the centralized repositories store employee data, such as demographics including age and gender, training records, performances evaluation, and career histories. The centralized systems support employee profiling by organizing data into structured profiles.

This study shows a digital solution for the Municipality of Meycauayan City that focuses on compiled employee profiles, attendance tracking, leave management, and performance monitoring to automate HR operations and improve public service delivery. According to **Kabir, M. et al (2021).** In the study of smart attendance and leave management system using fingerprint recognition for student and employees in academic institute. Attendance is an important factor in determining eligibility, punctuality, and commitment to the school. Maintaining and successfully managing attendance records is critical for grading students and staff. In the modern period, attendance is recorded on paper. In educational institutions, a system is in place that allows the teacher to declare the roll numbers of the students. This is a manual technique that is prone to human error and requires a significant amount of time for both students and teachers to complete the lessons.

Similarly, keep track of employee attendance. Keeping daily records and preparing monthly/annual reports using manual computations has become a significant undertaking.

Automated Leave Request

The automated leave request simplifies the process of managing employee absences, including vacations, sick days, or other time off, by replacing manual methods. The benefits of integration of automated leave request in the system is to enhances the operational continuity and resource allocations in the departments. This approach can result to improvement efficiency and time savings by eliminating manual process, allowing employee to submit request via self-service portals to administrator portals to approve the request quickly in the system to speed up the process. The adoption of automated leave request system is not only optimized daily operations but also position the organization to success by promoting more agile and convenient process. This approach will help the human resources department with the implementation and integration of automated leave request for municipality of meycauayan city. According to **Chowdhury, S.S. (2015).** In the study of Human Resources Automated leave request management: A Decision Support System. Address the inefficiencies in manual leave systems, including the paperwork delays. The study stated that by implementing and integration of automated platform the system enables employee to submit leave request online, generate reports on employee availability, absences monitoring and the approver to and disapprove the request of employee. The study also

addresses the key benefits include reduced processing time, minimizing the errors and real-time data visibility, enhancing the policy compliances and cost-effective management like corporate company

Automated Recruitment

The implementation and Integration of automated recruitment to the system simplifies the process of recruitment of applicant to choose accurately the qualified applicant for the open position and fair procedure. According to **Archibald, M. M. et al (2019).** In the study of using zoom videoconferencing for qualitative data collection: Perception and experiences of researchers and participants. The study is exploring the feasibility and acceptability of using zoom videoconferencing platform for collecting qualitative data. This study examines how the zoom's features make it suitable for conducting interviews remotely, particularly in research and recruitment context. This study also provides insights into using third-party video conferencing for interview purpose. According to **Alzharami, A.M. (2020).** In the study of the effectiveness of E-Recruitment Software over other Online-Based Recruitment Methods. This study assesses the effectiveness' of e-recruitment applications by focusing on the features these platforms offer to employers to make recruitment more productive. The study is comparing the e-recruitment applications focusing on job posting features and recruitment process effectiveness across different online platforms. This is study is also applicable the use of third-party software applications, because the proposed recruitment

process both use the third-party software application in job posting and online interview.

Performance Evaluation

The implementation and integration of performance evaluation to the system is to evaluate the process of employee performances and growth to enhance the work effectiveness and competitiveness. According to **Ishizaka A., et al. (2016)** In the study of portraying an employee performances management system based on the multi-criteria decision analysis and visual techniques. This study addresses the limitations of traditional performances appraisal systems, which often suffers from subjectivity, bias, and lack of clarity, by developing a structured frameworks that combines quantitative decision-making tools with visual aids to enhances decision-making in human resources management system. This study also directly relates to the implementation and integration of the performance evaluation to the system, it is providing the system-based approach to develop decision makings and visual analytics.

2.2 Hardware Literature and Studies

Fingerprint Scanner (R307)

Fingerprint Scanners R307 is a widely used optical fingerprint sensor for biometric authentication hardware to many application platforms for its robust applications in the history of biometrics. The fingerprint scanners are known for unique and secure and convenient way to identify and authenticate individuals, high accuracy and difficult to duplicate. According to **Oloyede, M. O et al (2013).** In the study of fingerprint biometric authentication for enhancing employee attendances system addressing the issues like proxy attendance. It focuses on employee profiling through fingerprint minutiae extraction, storing the biometric data into a centralized database to authenticate the accountability and generate attendance reports. This is highly recommended for human resources project like human resources information management (HRIS) system that aims to enhance accountability, accuracy and efficiency of data in workforces' management.

Arduino Uno

Arduino Uno is an open-source platform used for building electronics projects consist of both hardware and software combined that makes it easy to program and control devices. According to **Muchtar M. A. et al (2018).** In the research, Attendances fingerprint identification system using Arduino and single board computer. The study is focus on designs and implementation of biometric attendance identification system for efficient human resources management, suited for educational and workplaces settings. The objective of this study stated to create a secure, automated attendances system to replaces manual methods, reducing errors like proxy attendance and improving human resources process.

A4TECH PK-925H FHD Web Camera

A4TECH PK-925H FHD Web Camera is one of the components that use in the automated recruitment process to support for the online initial interview of human resource to applicant. According **Tincup W., et al (2025, July 17) I**n the study of web interviews improve your recruitment workflow. This study is derived from a podcast discussion and how explore how webcam-based interviews enhances recruitment workflows. This focus on integration of live and pre-recorded video interviews using web camera technology, emphasizing their role in initial interviews for hiring processes. In this study the author discusses practical benefits, challenges, and best practices for organization adopting webcam interviews. This study provides a practical, industry focused perspective on how webcam technology is implemented in modern hiring that covers both live and asynchronous formats.

2.3 Technical Background (Software Literature and Studies)

Figma

The system will start on creating design and prototyping phase to plan the design of the systems, before starting to developing phase. **Figma** is a collaborative design platform used for creating web interfaces (UI), user experiences (UX), prototype and software systems. According to **Frick, Albin. (2022)**. **From design to code:** A study on generating production code from user interface design software. The study is to investigate the tools like Figma for generating the code from UI prototypes, analyzing the prototyping phase where the design tested before actual software development, that focus on bridging the gap between designers and developers.

C++

C++ is one of the software language technologies we are using to build the fingerprint scanners integrates to the attendances of the system. According to the **Prabowo, N. K et al (2013).** The implementation of Arduino microcontroller boards in sciences. The biometrics analysis of 1122 Scopus papers (2008-2022) on Arduino in science. The Arduino IDE uses C++ for writing, uploading and debugging code to control sensors, IoT, data acquisition in projects like environmental monitoring. This study C++ can enable prototyping low-cost devices

and challenges include the code errors, and manual assembly. This study aims to provide a comprehensive understanding of C++ by the implementation and integration of ideas from credible sources to build skills in C++ on to integrate the C++ code to the hardware component, problem solving, circuitry.

Node.js

NodeJS is one of the software technologies we are using to build for the Human Resources Employee Management System for the Municipality of Meycauayan City is the node.js. This software technology helps to the faster development of the system. From the **developer.mozilla.org (n.d.),** Node.js is a cross-platform JavaScript runtime environment that allows developers to build server-side and network applications with JavaScript. This Technology is ideal for the system development to matches the requirements and handle multiple users, real-time updates, and document handling that creates a powerful and efficient solution.

ReactJS

ReactJS will serve as the frontend of the development of the system. This is a JavaScript library developed for creating user interfaces (UIs) in web development. Its core strengths lie in its component-based design and efficient rendering mechanism. It was emerged as a critical component in modern web development for building dynamic and interactive user experiences. According to

Mal N., et al (2024). The study of examine react.js concept evolution, key features, best practices, and performances optimizations drawing from credible sources to aid developers, educators and researchers. this article examines React.js in depth, covering its underlying ideas, evolution, key features, best practices, and performance optimization methodologies. This study aims to provide a comprehensive understanding of React.js by combining ideas from credible sources and industry expertise, thereby assisting developers, educators, and researchers.

MySQL

From the **MySQL.com (n.d)** is an open-sources relational database management system (RDBMS). widely used for storing, organizing, and managing structured data, particularly in web applications and other data-driven systems. MySQL is known for its widespread, reliability and ability to handle large volumes of data and high traffic, and it's very suitable for both small and large-scale applications. MySQL is compatible in programming frameworks and library like Laravel, React.js and another bunch of JavaScript language and also this is compatible to server-side language like PHP.

HTML (Hypertext Markup Languages)

HTML is widely used for creating web pages, and it also combines with different JavaScript libraries and the Laravel framework for well-structured output, such as dynamic web pages and faster development. From the **developer.mozilla.org (n.d.),** creating modern, functional, and user-friendly websites. HTML is the foundation of building website structure, which is paired with CSS for styling the web pages and JavaScript for interactivity, resulting in dynamic web pages.

Tailwind CSS

Tailwind CSS Is used in a utility-first CSS framework designed for rapidly building custom user interfaces directly within html. Unlike traditional UI frameworks Bootstrap or Material UI, which provide pre-designed components, Tailwind CSS offers a comprehensive set of low-level utility classes. From **Tailwind CSS (n.d.)** States the Tailwind CSS utility-first CSS framework designed for rapidly building custom user interfaces directly within HTML markup, without the need to write extensive custom CSS code. Make it easy to implement and integrate to the embedded HTML code with the use of class, resulting to clean, efficient and well structure code to enhanced user experience.

2.4 Integration of Software

The Node.js, React, JavaScript, HTML and Tailwind CSS will be the main technology to create the structure and architecture of system. That can work together to build the dynamic web application and seamless user experience and efficient system performances. The Node.js is will serve as the backbone of the server-side architecture, handling request, implementing and managing communication between frontend and database to ensures the efficient handling of user request and real-time data processing in the system. The React is serves as the system frontend technology, enable the creation of dynamic, responsive user interfaces through its component- based architecture. The JavaScript acts as the primary programing languages that use in the system across both frontend and backend environment, providing consistency in the development of the system. The MySQL serve as the database of the system to ensure the data integrity and manipulate information displayed in the user interface. The HTML serves as the skeleton and structure of web pages to define what appear on the pages. The Tailwind CSS will provide utility first styling, enable UI development through pre-defined classes to the system. This ensures the design consistency, responsive layouts a maintainable style without overhead of custom CSS files. With the integration of these technology to build the system will provide a solid foundation for the researchers to build a scalable and maintainable web application for the Human Resources Department of Municipality of Meycauayan City.

2.5 Integration of Hardware

The system hardware is focus on the integration of fingerprint scanners as biometrics in attendances. This integration is widely used in different platforms application systems, which is capable to handle attendances system for better record and employee monitoring.

The system will integrate the fingerprint scanners for attendances of employee to reduce reliance on manual and less reliance on third party software. The system hardware component is including fingerprint sensors serves as the biometrics data collection and converted into a template. The Arduino to handle the fingerprint sensor and communication. To save the template of the fingerprint and the attendances records and stored it to the database. The internet connection connectivity is important to transfer the data to the server and power supply to power up the scanner and controller.

With the use of these hardware, the operation of Municipality of Meycauayan City is expected to be improve.

2.6 Minimum and Recommended Specifications (PCs and Laptop)

The minimum computer specifications will suffice to operate the system solely for demonstrating its functionality. It is inadequate for managing the system over time, including the functioning of various processes.

Table 2.1: Minimum computer specification for End user

Components	Specifications
Processor	Intel Core i3 series (8th gen to latest)
RAM	8GB RAM
Storage	256 GB SSD or NVMe SSD
Graphics	Integrated Graphics
Operating System	Windows 11
Network	LAN based Internet Connection

Source: https://www.faceofit.com

The minimum specifications given by researchers are derived from the analysis and system requirements that guarantee to smooth system operation. The

Intel Core i3 processors, ranging from the 8th generation to the most recent generation, manage the system operations. The 8 GB of RAM (Random Access Memory) will enable seamless multitasking and quicker system responses tailored to the end user's requirements, avoiding slower reactions that may lead to undesirable errors during operation. The SSD Sata (Solid State Drive) or NVMe SSD offering 256 GB of storage will provide sufficient capacity for end users. For visualization, Intel's Integrated Graphics is sufficient for basic display tasks, and with optimizations from Windows 10 or 11, it will provide stability and compatibility. Internet access through service providers is sufficient to manage processes and connect to the system via the web.

Recommended specification for computer is advisable as it will ensure the operation of the system to its best performance for the long run, delivering a smooth performance of overall functionalities of the system.

Table 2.2: Recommended computer specification for End Users

Components	Specifications
Processor	Intel Core i5 (10TH Gen to Latest)
RAM	16GB RAM DDR4
Storage	1TB GB SSD or NVMe SSD
Graphics	Graphics card (8GB VRAM)
Operating System	Windows 11
Network	LAN based Internet Connection

Source: https://www.faceofit.com

In compliance with this recommended specification, the system is expected to perform better than the minimum specification. Offering a smooth workflow and minimizing loading time for data processing and dissemination. The Intel Core i5 (10TH Gen to Current) is assured to deliver the expected system performance, with the capability to handle stressful tasks and multitasking. The 16 GB RAM (Random Access Memory) will support the processing of larger data sets and multiple processes smoothly, in addition to a 1 TB storage drive capable of storing a huge volume of data, preventing sudden storage capacity crises while operating for prolonged periods. There will be a dedicated graphics card with 8 GB VRAM (Video Random Access Memory) to handle the display and intricate visualizations smoothly, resulting in an interface that is easier to use. Operating Systems such as Windows 10 or 11 will operate appropriately, as the operating systems are pre-

equipped with modern development tools and thus a good choice for the recommended system specifications. For improved registration of data and faster transfer speeds, internet service providers' high-speed connection provides faster access and enhanced responsiveness to continuing workflows.

Recommended components and specifications for the system are advisable, as they will ensure the operation of the attendances in the system to its best performance for the long run, delivering a smooth performance of the overall functionalities of the system.

Table 2.3: Hardware Component and specification

Components	Specifications

Fingerprint Sensor	Fingerprint Sensor R307: Type: Optical; Resolution: 500 dpi; Capacity: 1000 templates; Interface: UART; Supply Voltage: DC 3.6–6.0 V
Arduino	Arduino Uno R3 Microcontroller: ATmega328P; Operating Voltage: 5V; Flash memory: 32kb, Clock Speed 16 MHz
Power Supply	5V DC Regulated Adapter, 2A minimum
JST connector wire cable	JST connector wire cable 6 pin
A4TECH PK-925H FHD Web Cam	1920 × 1080 pixels compatible: windows 11 Frame rate: ~30 fps
Network	LAN based Internet Connection

Source: https://www.google.com

The Hardware Requirements including the component and specifications given by the researchers are expected to boost the attendance management in the system by implementing the fingerprint scanner. The integration of fingerprint

scanner is providing many technical, and security benefits compared to traditional method. In eliminates manual error and ensures each record is tied to a unique individual and automates the data collection. The addition and integration of web camera to the system is to help the human resources for conducting initial interview in the system using automated recruitment. This integration provides real-time monitoring and create reliable modern attendance and recruitment solution. This integration expected to improve the efficiency, reduce administrative cost and enhances overall accountability.

2.7 Theoretical Framework

This research is based on the Socio-Technical Systems Theory, Human Resource Information System (HRIS) Theory, and the Technology Acceptance Model (TAM), which serve as a basis for comprehending the incorporation of biometric technology into human resource management systems. The Socio-Technical Systems Theory (Trist & Bamforth, 1951) highlights that the successful implementation of technology relies on the interplay between social elements such as employees and human resource policies, and technical components like

fingerprint scanners, databases, and software. In the context of the Municipality of Meycauayan City, this theory reinforces the notion that the human resource employee management system should not only implement new technologies but also synchronize them with organizational processes and workforce requirements to enhance attendance monitoring, data management, and recruitment. The Technology Acceptance Model (TAM) proposed by Davis (1989) elucidates the acceptance of technology by users based on perceived usefulness and perceived ease of use. This model is pertinent as employees and human resource personnel must view the new biometric system as both beneficial in enhancing efficiency and data accuracy, and user-friendly with reliable fingerprint scanning to ensure successful adoption. These theoretical frameworks inform the design and execution of the Web-Based Human Resource Employee Management System utilizing Fingerprint Scanner technology. By examining the relationship between technology, users, and organizational structure, the study seeks to guarantee that the system is not only technically viable but also socially embraced and operationally sustainable. The Human Resource Information System (HRIS) Theory (Kavanagh & Johnson, 2017) centers on the integration of information technology within human resource functions to improve efficiency, accuracy, and decision-making. This theory posits that digital systems can facilitate HR activities such as employee profiling, attendance tracking, leave management, and recruitment by centralizing data and automating routine tasks.

2.8 Synthesis

This study is aim to provide a better user experiences to the end user that will used the systems in the future. The **Web-Based Human Resources Employee Management System using Fingerprint Scanner for the Municipality of Meycauayan City** is the transition can lead to the improvements and new workflow of the department, by implementing the automated systems that give better user experiences and improve overall performances for the end user. The systems ensure the capability of handle the data properly and securely using the centralized record in the database and display the employee information in the systems using employee profiling. The implementation of automated attendance system that can lead to better attendances monitoring, automated leave request to simplifies the process of managing employee absences, including vacations, sick days, or other time off, by replacing manual methods, employee performances evaluation, accurate process of evaluating employee performances and growth to enhanced work effectiveness and competitiveness. Accurate evaluation process of applicant using automated recruitment process by implementing fair process of choosing the right applicant that qualified into an open position. The implementation of the right software technology is providing better foundation on building the systems. Starting from design and prototyping phase to development phase that leading to fast the development. The integration of hardware in the system is design to connect to the attendance in the systems to better employee monitoring and to perform compatibility to both PCs and Laptop and seamless handling of operations such as real-time updates, data processing and performances evaluation. This result to accurate, efficient and way more effective

implementation of the system that give better user experience from day-to-day operations of the department. The proposed system differs from the similar existing by providing enhanced better user experience, improved functionality, and a more simplified workflow that makes it easier and more intuitive for user to navigates. In the proposed system, the improvement of the existing operations within the human resources department is emphasized. The included to the enhancement is the attendances, employee management, filing leave request, performance evaluation and also the recruitment process of the department. The system uniqueness lies in its comprehensive integration of the human resources functions within a simplified, user-friendly interfaces that prioritizes ease of use while maintaining robust functionality addressing the common pain point of overly complex existing human resource system.

The key innovations and functionality of the systems includes timekeeping which include attendances that integrates to fingerprint scanners and the calculation of work hours, tardiness, undertime, daily time records and corrections. The timekeeping has a comprehensive attendance report with customizable date ranges and filtering options and also the anomaly detection system identifying suspicious attendances patterns or irregularities and lastly it will send instant notifications to administrator. The automated leave request workflow with simple approval routing, real-time leave credit balance with automatic calculations, calendar visualizations displaying department leave schedules to prevent conflict. The automated recruitment includes initial interview scheduling automation with calendar integration and email verifications, document verifications and credential

checking workflow. This ensures fair and transparent applicant evaluation in initial online interview. The performance evaluations include organizational objectives, predefined metrics and competencies, historical performance comparison and trend analysis for career development planning and competency gap identification with training recommendation. This kind of process target the accuracy and efficiency to evaluate the employee performances and growth resulting to work effectiveness and competitiveness. Reporting and analytics dashboard includes import and export functionalities in multiple formats using pdf and csv forms for both internal and external reporting, compliances reporting aligned with civil service commission requirements and data visualization through interactive charts and graph analyses. The employee self-service portal including personal information updates with approval workflow ensuring data accuracy, leave applications submission with real-time status, access to performances reviews and acknowledgment features and document repositories for employment records, certificates and personal files. The enhanced security and access control includes role-based access control (RBAC) with customizable permission level and automated backup mechanisms ensuring data integrity and disaster recovery. The scalability and future integration readiness includes modular architecture design to accommodate future enhancement and additional modules, API capabilities enabling potential integrations with payroll systems and other municipal databases, configurable workflows adaptable to evolving departmental policies and procedures and cloud-ready infrastructure for potential migrations to cloud-based hosting. These innovations address the limitations of traditional human

resource management symbols by providing a comprehensive, integrated solution to operation, reduces administrative burdens, improves data accuracy, enhances decision making capabilities through real-time analytics and reporting

Although various literature and studies affirm the system's potential, further research is essential to address existing gaps and fully maximize the effectiveness of the Human Resources Employee Management System.

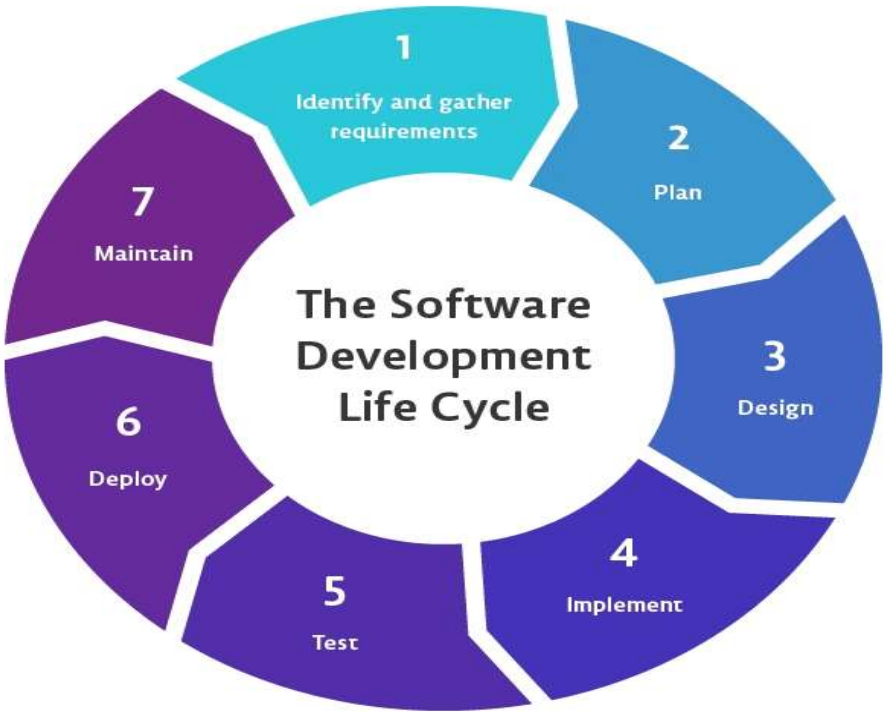
CHAPTER 3

RESEARCH METHODOLOGY

This chapter will discuss the research methodology of the proposed Web Based Human Resources Employee Management System using Fingerprint

Scanner for Municipality of Meycauayan City which involves the research design, the quantitative method, along with statistical techniques, research procedure, system architecture and research instrument employed. This method helped the researchers and develop a Fingerprint scanner and Web-Based system.

Research Design



The study makes use of System Development Life Cycle (SDLC) Methodology, SDLC was selected as it gave a structured and well-organized method for system development which made sure that every step from the planning, analysis, design and implementation to the testing and maintenance departments was carried out properly. This methodology was applied in developing

the Web-Based Human Resources Employee Management System using Fingerprint Scanner for the Municipality of Meycauayan City, to enhance the efficiency of the system, its reliability and alignment with the organization's demands is necessary. To evaluate the effectiveness of the proposed system, the study utilized a Quantitative Research Method through the use of structured survey questionnaires.

The method was chosen because it produces physically and objectively-measurable data and can be statistically analyzed to demonstrate the effectiveness of system. The data gathered was used to rate the usability of the system and accuracy, reliability, efficiency, and user satisfaction. The SDLC Methodology together with quantitative evaluation have guaranteed that the system not only achieved the necessary technical and functional requirements but also properly accommodated the timekeeping and human resource management demands of the Municipality of Meycauayan City.

DATA FLOW DIAGRAM

DFD Level 1 Diagram

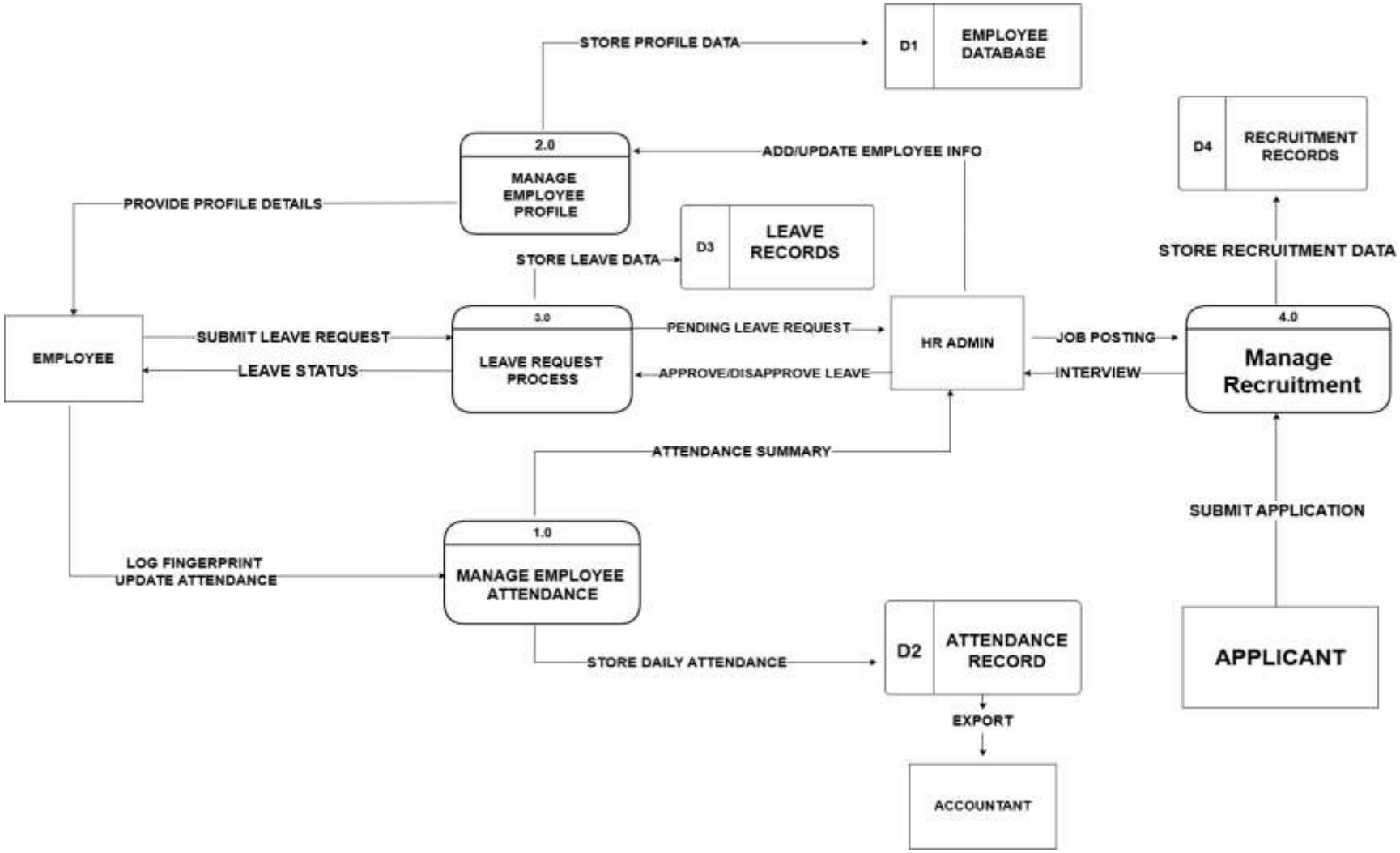


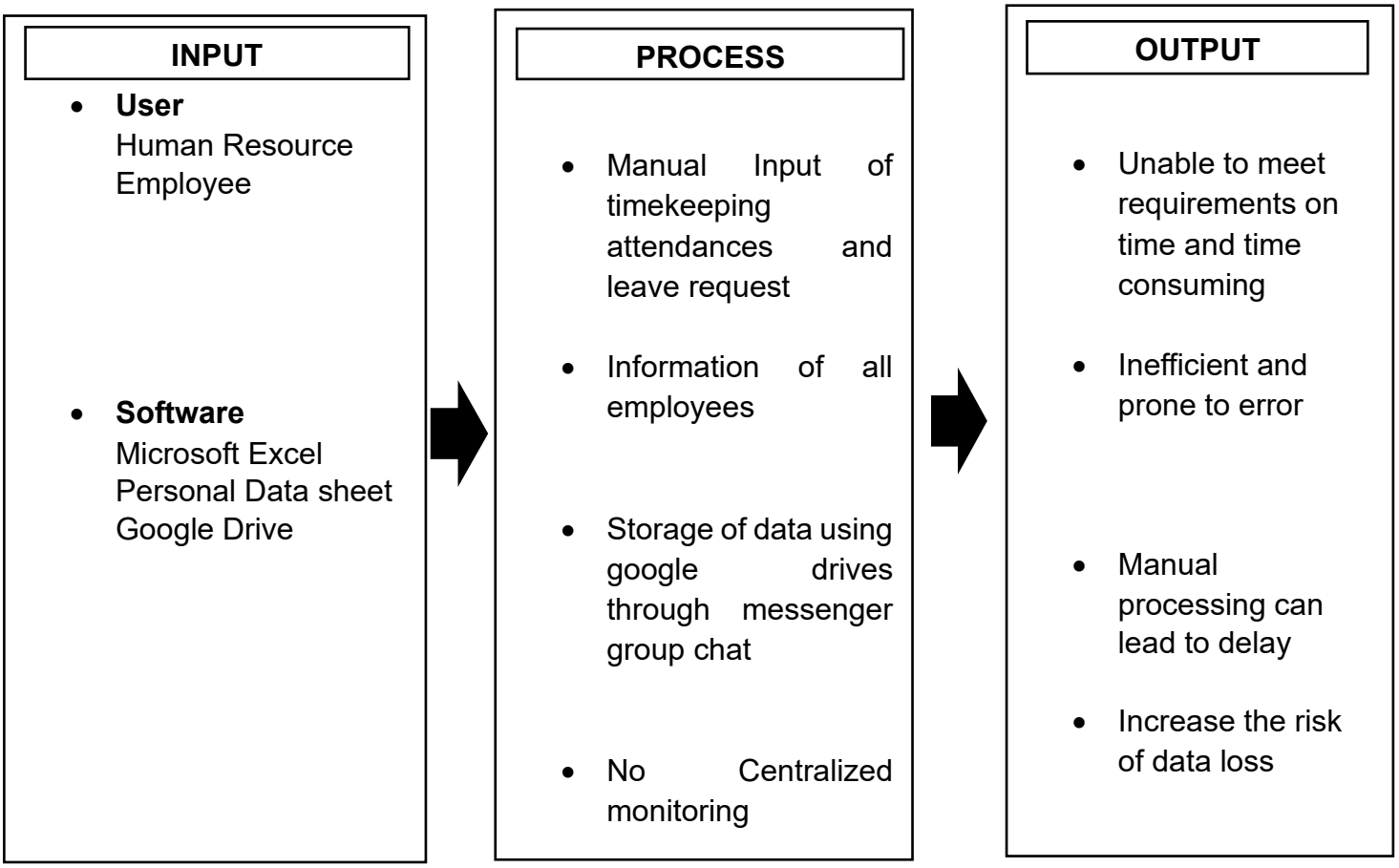
Figure 3.1 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram illustrates the detailed processes and data interactions within the Web-Based Human Resource Employee Management System Using Fingerprint Scanner. It shows how employees, HR administrators, and applicants communicate with the system to perform various HR-related functions such as attendance monitoring, employee profiling, leave processing, and recruitment management. The first process, Manage Employee Attendance (1.0), begins when employees log their attendance using the fingerprint scanner. The system automatically updates and stores this information in the Attendance Record (D2). The HR Admin can access attendance summaries to monitor employee performance and ensure attendance accuracy. The second process, Manage Employee Profile (2.0), allows employees to provide their personal and work-related details, which are then stored in the Employee Database (D1). The HR Admin can add, update, or modify employee information to maintain accurate and up-to-date records within the system. The third process, Process Leave Request (3.0), manages the submission and approval of employee leave requests. Employees can submit their leave applications through the system, which forwards them to the HR Admin for review. The HR Admin evaluates each request and either approves or disapproves it. The decision is then communicated back to the employee as a leave status update, and all related data is recorded in the Leave Records (D3). The fourth process, Manage Recruitment (4.0), focuses on the hiring process. The HR Admin can post

available job openings, and applicants can submit their applications directly through the system. Once received, the HR Admin reviews the applications, schedules interviews, and stores all recruitment-related data in the Recruitment Records (D4) for documentation and tracking purposes. Overall, the DFD Level 1 diagram demonstrates how the system efficiently integrates and automates major HR functions. By connecting employees, HR administrators, and applicants through a centralized platform, it ensures accurate recordkeeping, faster HR transactions, and a more organized workflow within the organization

Conceptual Framework

Existing



The Existing System shows the current operations of the HR department. Heavily reliance's on manual input of data, documentation and timekeeping using physical logbooks and third-party software can lead to inefficient and inaccuracy way of operation.

Figure 3.2 Conceptual Framework of the Existing System

Proposed

- **User**
  Human Resource
  Employee
- **Software**
  HTML, Tailwind CSS, ReactJS, NodeJS
  C++
- **Hardware**
  Fingerprint Scanners
- **Database**
  MySQL
- **Technology**
  VS code, Laragon, Microsoft Edges

- Automated and digitalized process of monitoring of employee attendances
- Centralized process of employee data management record and Storage
- Simplified the process filing leave request
- Accurate process to evaluate employee performances and growth
- Accurate evaluation of applicant using automated recruitment

- Reliable and precise records of employee hours worked and increased the transparency
- Accuracy and efficiency storing the data in one centralized database
- Accurate managing of leave request using automated leave request
- Enhance work effectiveness and competitiveness
- Accuracy of choosing qualified applicant into an open position in fair procedure

Figure 3.3 Conceptual Framework of the Proposed System

The proposed system, shows the human resources capability to less the process of relying in the manual process and third-party software, although the system lessen to the third-party software applications, will continue to use in enhances way for exporting and importing file. The system features an automated system that provide more accurate and efficient way of operation for the Municipality of Meycauayan. The system includes, timekeeping to process and monitor employee attendances using fingerprint scanner and result to better worked and attendances transparency. Employee Profiling provides accuracy of employee data in one centralized database. The process of applying for leave requests has been simplified using an automated method. The employee performances evaluation provides an accurate and efficient evaluation of every employee for better effectiveness and competitiveness. The recruitment process ensures accuracy and provides assurances for selecting the most qualified applicant for an open position, adhering to fair hiring procedures.

Research Locale and Population

The study was aimed at Municipality of Meycauayan, located in Meycauayan City, Bulacan. The company was selected as the location for conducting research locale because it currently uses manual and third-party

application of timekeeping, employee profiling, leave request, applicant recruitment, and employee performance evaluation which the main purpose of proposed system is to improve.

1. **Human Resource Admin** - The responsible for managing timekeeping, employee profiling, leave requests, recruitment, and performance evaluation. They help identify system inefficiencies and assess how the proposed system can enhance accuracy and minimize manual tasks.

2. **Employees** - especially those in the employee recruiting and profiling, will use a new automated web-based employee management system that includes fingerprint scanner. This system will improve the organization and managing records that making retrieval more efficient.

3. **Applicant–**The applicant serves as a user of the system during the recruitment process. They can submit their personal information, credentials, and application forms through the web-based system. The system allows them to track their application status, receive notifications for interviews or requirements, and ensures their data is securely stored and efficiently managed by the HR department.

4. **Owner-** The owner supervises the complete installation and administration of the outlined system for the Municipality of Meycauayan. The owners ensure

that the system is responsive to the organization's needs, follows rules, and adheres to policies regarding the management of data. They also approve resource allocation and oversee the integration of the system throughout the organization and assess overall impacts to employee performance and productivity. Owner feedback is a critical component in assessing the system's impact on operational efficiency.

Sampling Method Slovin Formula:

$$n = \frac{N}{1 + Ne^2}$$

Where:

N = population size

e = margin of error (0.05)

n = sample size (expected respondents)

Computation:

$$n = \frac{18}{1 + 18(0.05)^2}$$

$$n = \frac{18}{1 + 18(0.0025)}$$

$$n = \frac{18}{1 + 0.045}$$

$$n = \frac{18}{1.045}$$

Sample Size (n) = 18 respondents

The proponents used **simple random sampling** together with **Slovin's formula** to determine the appropriate sample size for the study. This method ensured that every member of the target population had an equal chance of being selected, providing accurate and unbiased data for evaluating the proposed system.

Description of the Respondents

The respondents of this study are the HR personnel, supervisors, and employees of **"A Web-Based Human Resources Employee Management System Using Fingerprint Scanner for Municipality of Meycauayan City"**, who were purposely selected because they are directly involved in leave requests, employee evaluation, and recruitment**.**

The human resource personnel evaluated the accuracy and efficiency of the payroll module, the supervisors assessed the scheduling features such as workload balancing and conflict resolution, while the employees focused on the usability and reliability of the self-service portal. Their feedback ensured that the evaluation of the proposed system reflected the perspectives of its actual end-users, making the results relevant and aligned with the organization's operational needs.

Table 3.1 Description of the Respondents

Population	Number of Respondents
Human Resource	1
Administrative	8
Senior Administrative	2
Employees	7
Total Respondents	18

Research Instrument

The study used a survey questionnaire as its main research instrument. The questionnaire was designed using a Likert scale with response options of Satisfied, Neutral, and Dissatisfied, focusing on important criteria to evaluate the effectiveness of the proposed system. These criteria included user-friendliness, reliability, efficiency, transparency, and accuracy in handling payroll, attendance, and scheduling processes. The feedback collected from Human Resources personnel, supervisors, and employees provided measurable insights into how well the system addressed their needs. The data gathered from the questionnaires were analyzed and presented in the Results and Discussion section, where they were linked back to the Statement of the Problem to determine if the system met the objectives of the study and resolved the identified issues.

Questionnaire Design

A Web-Based Human Resources Employee Management System Using Fingerprint Scanner for Municipality of Meycauayan City

Name:	**Department:**
Position:	

Instructions: Please check and indicate your level of agreement with the following statements by selecting and response for each item.

1 – Strongly Disagree **2** – Disagree **3** – Neutral

4 – Agree **5** – Strongly Agree

Table 3.2 Description of the Respondents

ITEMS TO BE EVALUATED	**EXISTING SYSTEM**						**PROPOSED SYSTEM**				
	1	2	3	4	5		1	2	3	4	5
1. Does the system meet the operational needs of HR Department?											
2. The existing/proposed recruiting system is accurate and dependable?											
3. The existing/proposed attendance monitoring system frequently causes delays in processing?											
4. The organized system improves data accuracy and decision-making.											

Figure 3.4 Questionnaire Design

Validation and Establishing Reliability

The research instrument utilized in this study was a researcher-made questionnaire specifically developed to evaluate the effectiveness of the proposed system. To ensure its validity, the instrument was presented to the research adviser and subject matter experts for content evaluation. The validators examined the clarity, relevance, and alignment of the items with the study objectives. Based on their feedback, necessary revisions were made, after which the final version of the instrument was approved for administration. Furthermore, the instrument was assessed for consistency and was found to be reliable for collecting the required data.

Statistical Treatment

The data gathered from the respondents were analyzed using descriptive statistics. The responses from the survey questionnaires, which used a 5-point Likert Scale, were tabulated and interpreted through the weighted mean to determine the overall assessment of the system in terms of efficiency, accuracy, usability, and productivity. The standard deviation was also computed to measure the consistency of the respondents' answers.

The mean (average) was applied to evaluate the responses from the survey questionnaires, as it provides a straightforward way to summarize the general sentiment of the respondents. By computing the average score for each item, the

mean offers a clear indication of opinions on the system's effectiveness, usability, accuracy, and reliability.

Mean Formula:

$$\text{Mean} = \frac{\Sigma\, xi}{N}$$

$$\text{Mean} = \frac{\text{Sum of All Data Points}}{\text{Number of Data Points}}$$

x

Cost Benefit Analysis

The Cost-Benefit Analysis (CBA) is a systematic process that evaluates the attractiveness of a decision by considering its possible advantages and the projected expenses associated with executing a project or system. This section outlines the cost assessment for software, hardware, and personnel necessary for the successful completion of the project within a specified timeframe.

System Cost

Table 3.3 Hardware Cost

Items	Qty	Unit Price	Total
Fingerprint Sensor R307	1	799	799
Arduino Uno R3 atmega328p	1	578	578
Power Adapter 12v 2A	1	495	495
A4Tech PK-925H FHD Web Cam	1	1050	1050
JST connector wire cable	5	113	399
Total Hardware Cost			**₱3321.00.**

Source: Shopee.com.ph

The table present the total hardware expenses associated with the proposed system. These components are essential for the development of the proponents' security of the system. The components listed include Fingerprint scanners R307 sensor, Arduino Uno R3 ATmega328P, Power Adapter, A4Tech PK-925H FHD Web Cam, and JST connector wire cable. The total cost for the aforementioned hardware components amounts to **₱3321.00.**

Table 3.4 Software Cost

Items	Annual	Total
Digital Ocean	**₱ 5,044.88**	**₱ 5,044.88**

Source: www.google.com

The table outlines the total software expenses associated with the proposed system. The proposed system will be hosted on digital ocean for a month, which incurs a cost of **₱5,044.00.** It is important to note that the web hosting provider only covers the software expenses, resulting in a total software cost of **₱ 5,044.00**

Table 3.5 Professional Cost

Personnel	Duration	Professional Cost	Total
Software Developer	6 months	₱85,000.00	₱85,00.00
System Design & Analysis	6 months	₱45,000.00	₱45,000.00
Hardware Developer	6 months	₱25,000.00	₱25,000.00
Total Professional Cost			**₱155,000.00**

The company offers several types of contracts designed to cover every stage of a project lifecycles. The **Software Developer Contract** for six months development is reaching almost **54.84%** percent of the contract. This Includes the development of the software system's frontend, backend, database and the deployment of the entire systems. The total cost allocated for the software developer is reaches to **₱85,000.00** in six months period. The **System Design & Analysis Contract** accounting for **29.03%** of the total six months project cost, is responsible for the design and implementation of the entire system. This includes the creation of the system architecture, workflow design, data structure planning, and the testing phase to ensure the system functions as intended. The role also covers performing various classes of testing, such as functionality, integration, and validation test to guarantee that the system meets both technical requirements and user's needs. The total cost of System Design & Analysis is reaches to **₱45,000.00** in six months development. The projects involve physical setup, the **Hardware Developer Contract** provides related to the devices assembly, installation, testing and integration within the system. This components of the project accounts for **₱25,000.00 or 16.13%**. The total professional cost for the proposed system is amounts to **₱155,000.00**

Table 3.6 Operational Cost

Details	Unit Price	Week	Months	Cost
Maintenance	**₱10,000**	**1**	**6**	**₱10,000.00**
Total Operational Cost				**₱10,000.00**

Source: upwork.com

The table present the operational cost that is applicable for the maintenances of the proposed system. This table are based from the source gathered from upwork.com one of the trusted sourcing platforms. The maintenances are included to maintaining the overall parts of the system, including the post-deployment system support and troubleshooting. The operational cost for the proposed system totals to **₱10,000.00**

Table 3.7 Project Investment Cost Summary

Description	Price
Hardware Cost	**₱3,321.00**
Software Cost	**₱5,044.88**
Professional Cost	**₱155,000.00**
Operational Cost	**₱10,000.00**
Contingency Cost	**₱34,673.176**
Total Project Investment Cost	**₱208,039.056**

The table provides a summary of the total investment cost for the proposed system project. It includes the total costs for hardware, software, professional services, and a contingency plan, which accounts for **20%** of the combined costs of hardware, software, professional and operational cost. The overall project investment cost amounts to ₱208,039.056

Table 3.8 Benefits Cost of the Owner

Benefits Category	Estimated Annual Benefit
Reduce Labor Cost	₱223,334.00
Reduce Material Cost	₱14,004.00
Total Annual Benefits	**₱237,348.00**

Benefits	**Minutes**	**Daily**	**Monthly**	**Annually**
Labor Benefits	37.6 minutes	₱846.00	₱18,612.00	₱223,344.00
Material Benefits		₱937.00	₱11,244.00	₱11,244.00
Total		₱1,783.00	₱29,856	**₱234,588.00**

The Reduced Labor Benefits were determined based on the time saved through the automation of HR processes such as attendance logging, leave filing,

document handling, and recruitment. Before the system was implemented, employees spent an average of **37.6** minutes per day performing these manual tasks. With a daily wage of **₱600** for an 8-hour workday (equivalent to **₱1.25** per minute), this means each employee saves about **₱47.00** per day. For **18** employees, the total savings reach **₱846.00** daily, which amounts to **₱18,612.00** monthly based on **22** working days. Over the course of one year, this results in a total labor benefit of **₱223,344.00**. The Reduced Material Cost Benefits were computed based on the decreased use of office materials such as bond paper and printer ink after shifting to a digital system. Previously, the HR department consumed approximately one ream of bond paper **₱230** and one printer ink bottle **₱707** per month, totaling **₱937.00** monthly. Through the implementation of digital forms and paperless transactions, this expense is largely eliminated, leading to an estimated **₱11,244.00** in annual supply savings. Overall, the proposed system provides a total of **₱234,588.00** in annual tangible benefits, demonstrating that it significantly reduces manual workload, saves processing time, and minimizes operating expenses within the Human Resource Department of the Municipality of Meycauayan City

Table 3.9 Project Investment Cost Summary Cost-Benefits Analysis

Year	Expected Benefits	Cost of Investment
1	**₱234,588.00**	**₱208,039.056**
Net Profit		**₱26,548.944**
Return of Investment (ROI)		**12.76%**
Payback Period		**10.64 months**.

The table presents the Cost-Benefit Analysis for the proposed project. It indicates that the total anticipated benefits of the proposed budget exceed the annual cost of the investment. The overall expected benefits amount to **₱234,588.00**, while the investment cost for one year is **₱208,039.056** These figures result in a net profit of **₱47,800.532**. The return on investment is **12.76%** thus, the payback period is **10.64 months**.

Cost-Benefits Analysis Formulas

Return of Investment Formula:

$$ROI = \left(\frac{Total\ Expected\ Benefits - Total\ Investment\ Cost}{Total\ Investment\ Cost} \right) x\ 100$$

$$ROI = \frac{(₱234,588.00 - ₱208,039.056)}{₱208,039.056}\ x\ 100$$

$$ROI = \frac{(26,548.944)}{₱208,039.056}$$

$$ROI = 0.12761519164\ x\ 100$$

$$\boxed{\boldsymbol{ROI} = \mathbf{12.76\%}}$$

Payback Period Formula:

$$pp = \left(\frac{Initial\ Investment}{Cash\ Flow\ per\ Year} \right) x\ 12$$

$$pp = \frac{(208,039.056)}{₱234,588.00}\ x\ 12$$

$pp = 0.8868273569\ x\ 12$

$\boldsymbol{pp} =$ **10.64 months**

The Return on Investment (ROI) was calculated by deducting the total anticipated benefits from the total investment cost, followed by dividing the resulting difference by the total investment cost. The resulting quotient is then multiplied by **100** to express the result as a percentage. In this instance, the calculation yielded an ROI of **12.76%**. The Payback Period (PP) was established by dividing the initial investment by the annual cash flow. This quotient is subsequently multiplied by **12**, resulting in a payback period of **10.64 months.**

Intangible Benefits:

1. **Strengthened Data Security:** The integration of fingerprint scanning enhances security by ensuring that only authorized personnel can log in or record attendance, reducing risks of identity fraud or "buddy punching."

2. **Improved Accountability:** Biometric records create a transparent and tamper-proof log of employee attendance, encouraging punctuality

and responsibility.

3. **Enhanced Communication:** The centralized web-based system allows real-time updates and notifications, improving coordination between HR and employees without the need for physical follow-ups.

4. **Boosted Employee Engagement:** Providing employees with easy access to their records, leave balances, and announcements fosters a sense of inclusion and involvement in company processes.

5. **Better Organizational Image:** Adopting modern HR technology demonstrates the company's commitment to innovation and efficiency, which can positively impact employee perception and attract potential.

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PROGRESS MONITORING

TITLE: "A WED-BASED HUMAN RESOURCES EMPLOYEE MANAGEMENT SYSTEM USING FINGERPRINT SCANNER FOR MUNICIPALITY OF MEYCAUAYAN CITY"

Hardware Adviser: Mr.Dailo Apollosz

Software Adviser: Mr.Dailo Apollos

Proofreader: Mr.Meljohn Navalez

CAPSTONE 1

Prepared By: Researcher

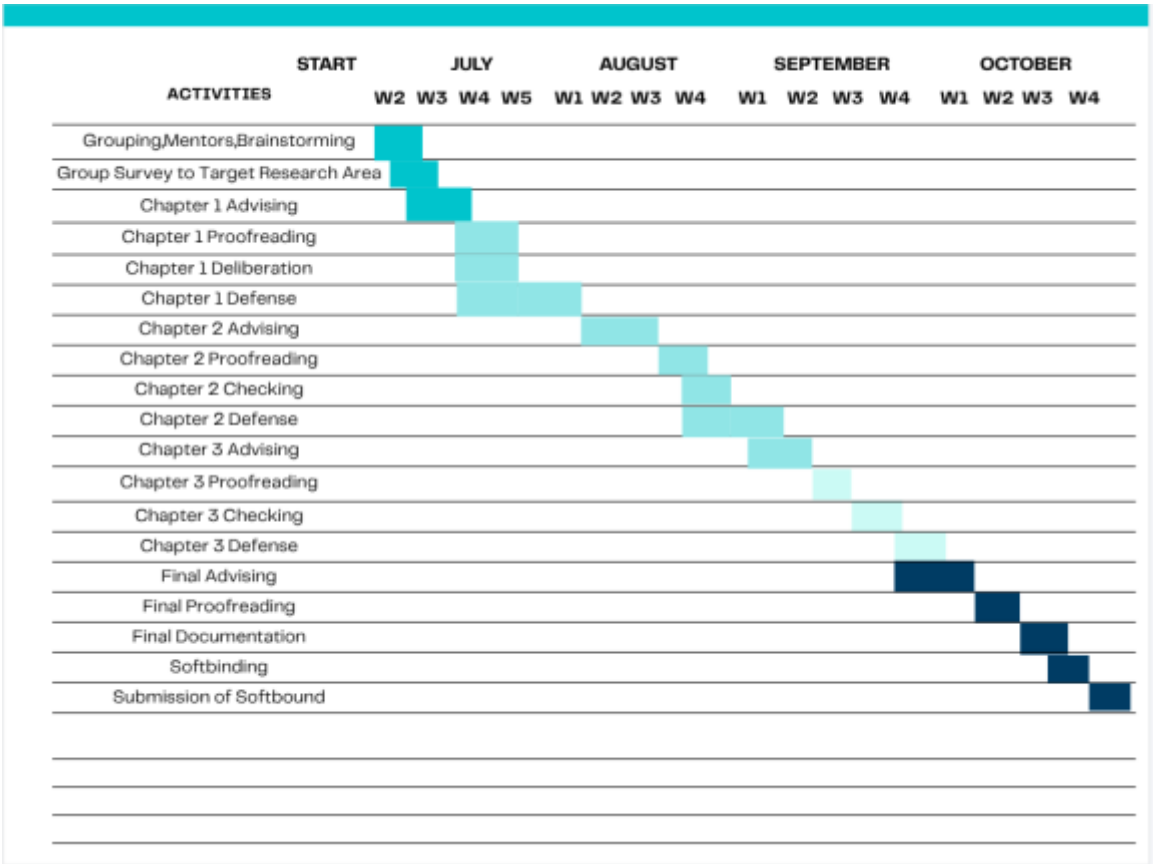
DATE	ACTIVITIES	MENTORS	SIGNATURE	REMARKS
	Groupings	Sir Nael		Completed
	Brainstorming	Sir Nael		Completed
	Target Industry survey	Sir Nael		Completed
	Title Project Idea	Sir Nael		Completed
	Mentor Identification	Sir Nael		Completed
	Chapter1			
	Advising	Sir Apollos		Completed
	Proofreading	Sir Meljohn		Completed
	Defense	Sir Nael		Completed
	Chapter 2			
	Brainstorming	Sir Nael		Completed
	Advising	Sir Apollos		Completed
	Proofreading	Sir Meljohn		Completed
	Defense	Sir Nael		Completed
	Chapter 3			
	Brainstorming	Sir Nael		Completed
	Advising	Sir Apollos		Completed
	Proofreading	Sir Meljohn		Completed
	Defense	Sir Nael		Completed

Camacho ,FredAllen G. Palero, Joshua P.

Bantigue,VictorLouie Somosa, sairel

Engr. Natanael Mendoza
IT Capstone Adviser

Gantt Chart


START
JULY
AUGUST
SEPTEMBER
OCTOBER
ACTIVITIES
W2 W3 W4 W5 W1 W2 W3 W4 W1 W2 W3 W4 W1 W2 W3 W4
Grouping,Mentors,Brainstorming
Group Survey to Target Research Area
Chapter 1 Advising
Chapter 1 Proofreading
Chapter 1 Deliberation
Chapter 1 Defense
Chapter 2 Advising
Chapter 2 Proofreading
Chapter 2 Checking
Chapter 2 Defense
Chapter 3 Advising
Chapter 3 Proofreading
Chapter 3 Checking
Chapter 3 Defense
Final Advising
Final Proofreading
Final Documentation
Softbinding
Submission of Softbound

FRED ALLEN G. CAMACHO

Sanville Bulak Street, Malhacan,
Meycauayan, Bulcan
Phone: 09274922359
Email: lpgallen04@gmail.com

OBJECTIVE

To gain valuable work experience in a professional environment, where i can apply my willingness to learn, dedication, and ability to work with a team.

SKILLS

- Basic computer knowledge
- Good Communication Skills
- Fas t Learner
- Can work under pressure
- Willing to be trained
- Maintains good personal hygiene
- Able to work well in a team

PERSONAL INFORMATION

Age: 21
Date of Birth: September 4 2003
Civil Status: Single
Nationality: Filipino
Religion: INC

WORK EXPERIENCE

Position: Food Pavillion Crew
Venue: Phillipine Arena, Cuidad de Victoria, Bocaue, Bulacan

Duties and Resposibilities

- Assisted customer in finding product
- Conducted daily inventory checks and reported stock levels
- Operated cash register and processed customer payment

EDUCATIONAL BACKGROUND

Elementary School
Malhacan Elementary School 2010 - 2016

Junior High School
A.F.G.B.M.T.S
2017 - 2020

Senior High School (TVL - Automotive)
A.F.G.B.M.T.S
2021 - 2022

College (Bachelor Of Science in Information Technology)
ACLC College of Meycauayan
2022 - 2025

Somosa Sairel

INFORMATION TECHNOLOGY

+63 999-560-0329 · relsomosa@gmail.com
Blk1A Lot 2 Cardinal Village Salitran 3 Dasmarinas Cavite

CERTIFICATION

19 January 2025	**Information Security (OpenLearn)**
17 April 2024	**Practical Password cracking Office files | Ethical Hacking (Udemy)**
17 April 2024	**Ethical Hacking:Command Injection (Udemy)**
17 April 2024	**Python Complete Course for Python Beginners(Udemy)**
29 July 2025	**Essential of Information Technology Technical Skills Tech support Level 1**
20 April 2024	**Cyber Security's Frontier: Emerging Trends and Technologies (Udemy)**

EDUCATION

Meycauayan, Bulacan Graduated - 2015	**Malhacan Elementary School**
Punturin Phase 5 Valenzuela NCR Graduated - 2019	**Vicente P. Trinidad National High School**
Banga Meycauayan Bulacan Graduated - 2022	**ACLC College of Meycauayan (SHS)**
Banga Meycauayan Bulacan	**ACLC College of Meycauayan (Current)**

SKILLS

Management Skills

Creativity

Digital Marketing

LANGUAGE

ENGLISH

FILIPINO

EXPERTISE

Cybersecurity Pentest

Network Analyst

Editing

Victor Louie Bantigue

INFORMATION TECHNOLOGY

09617903515 - louiebantigs@gmail.com
422 Muralla St. Libtong Meycauyan City Bulacan

PERSONAL INFORMATION

Age: 21
Date of Birth: March 24, 2004
Civil Status: Single
Nationality: Filipino
Religion: Roman Chatholic

SKILLS

- **Good Communication**
- **Adaptability**
- **Fast Learner**
- **Teamwork & Collaboration**
- **Good Listener**

WORK EXPERIENCE

Position: Barista, Kitchen Helper

Venue: Meycauayan City

Duties and Resposibilities

- **Inventory Management**
- **Cash Handling & POS**
- **Food Preparation**
- **Stock and Storage**

LANGUAGE

ENGLISH

FILIPINO

EDUCATIONAL BACKGROUND

General Academic Strand
Lawang Bato National High School
2019-2020

Our Lady of Fatima University Valenzuela
2020-2021

Systems Technology Institute (STI) Meycauayan
2022-2023

Bachelor Of Science in Information Technology
ACLC College of Meycauayan
2024 - 2025

Joshua P. Palero

INFORMATION TECHNOLOGY

09368077249 · joshuapalero111a@gmail.com
Meyland Homes Phase 2 Lawa Meycauyan City Bulacan

PERSONAL INFORMATION

Age: 24
Date of Birth: November 2 2001
Civil Status: Single
Nationality: Filipino
Religion: Roman Chatholic

SKILLS

- **Computer Programmer**
- **Web Development**
- **Teamwork Strong**
- **Work Ethics Good**
- **Listener**

WORK EXPERIENCE

Position: Software Developer
Marketing

Venue: ELI IT Solutions
Valenzuela City

Duties and Resposibilities

- **Internal Project Software Developer**
- **Frontend and Backend Development**

LANGUAGE

ENGLISH

FILIPINO

EDUCATIONAL BACKGROUND

General Academic Strand
Bitaogan National High School
2019-2020

Partido State University Goa-Caramoan Campus
2020-2021

Bachelor Of Science in Information Technology
ACLC College of Meycauayan 2022 - 2025

ACLC COLLEGE OF MEYCAUAYAN

Bachelor of Science in Information Technology

MC Mall Building, McArthur Hi-Way, Brgy. Bangcal, Meycauayan City, Bulacan, Philippines

Subject: Request for Permission to Conduct a Survey

Dear Ma'am Judith S. Guevarra,

Good day!

We are students of **Bachelor of Science in Information Technology** from **ACLC College of Meycauayan**, currently conducting our Capstone Project entitled
"Web-Based Human Resource Employee Management System with Fingerprint Scanner."

In line with this, we would like to request your permission to conduct a **survey** among the selected employees of your organization. The information gathered will serve as data for our research study and will be used solely for academic purposes.

We assure you that all responses will be kept strictly confidential, and the activity will not disrupt your regular work operations.

Thank you for your time and kind consideration. We are hoping for your favorable response.

Respectfully yours,
Bantigue, Victor Louie S.
Camacho, Fred Allen G.
Palero, Joshua P.
Somosa, Sairel
Researcher

Approved by:
Engr. Natanael Mendoza
IT Capstone I General Adviser

PRICELIST

Preferred High Quality R307 R307S Fingerprint Identification Module Sensor with Cable 15cm

4.0 ★★★★☆ 4 Ratings 14 Sold Report

₱799

Shipping: Guaranteed to get by 12 - 15 Oct
Get a ₱50 voucher if your order arrives late.

Shopping Guarantee: Free & Easy Returns

Quantity: 1 + 1 pieces available

Add To Cart | Buy Now

Mall Boon Technology Arduino Uno R3 Atmega328p Ch340g

5.0 ★★★★★ 1 Rating 4 Sold Report

₱578 ~~₱1,260~~

Shop Vouchers: ₱15 OFF ₱30 OFF ₱90 OFF ₱150 OFF ₱225 OFF Show All

Shipping: Guaranteed to get by 12 - 16 Oct
Get a ₱50 voucher if your order arrives late.

Shopping Guarantee: 100% Authentic · Free & Easy Returns

Chip Board: Atmega328P | CH340G

Cable: with USB B | without USB B

Quantity: 1 + 23 pieces available

Power Adapter 5/6/12V 0.5/1/2/3/4/5A 150CM Cable Plug Size 5521

4.9 ★★★★★ | 136 Ratings | 3K+ Sold | Report

₱159 - ₱459 ~~₱344 - ₱561~~ -54%

Shipping: Guaranteed to get by 18 Oct >
Free shipping
Get a ₱50 voucher if your order arrives late.

Shopping Guarantee: 100% Authentic · Free & Easy Returns · Electronic Protection

Power: 5V 1A | 5V 2A | 5V 3A | 6V 1A | 9V 2A | 12V 1A | 12V 1.5A | 12V 2A | 12V 3A | 12V 4A | 12V 5A

Quantity: 1 IN STOCK

5Pcs VH3.96 VH 3.96mm JST Wire Female Housing Plug Connector Wire Cable 2 3 4 5 6Pin 20AWG Wire 10cm 20cm Length

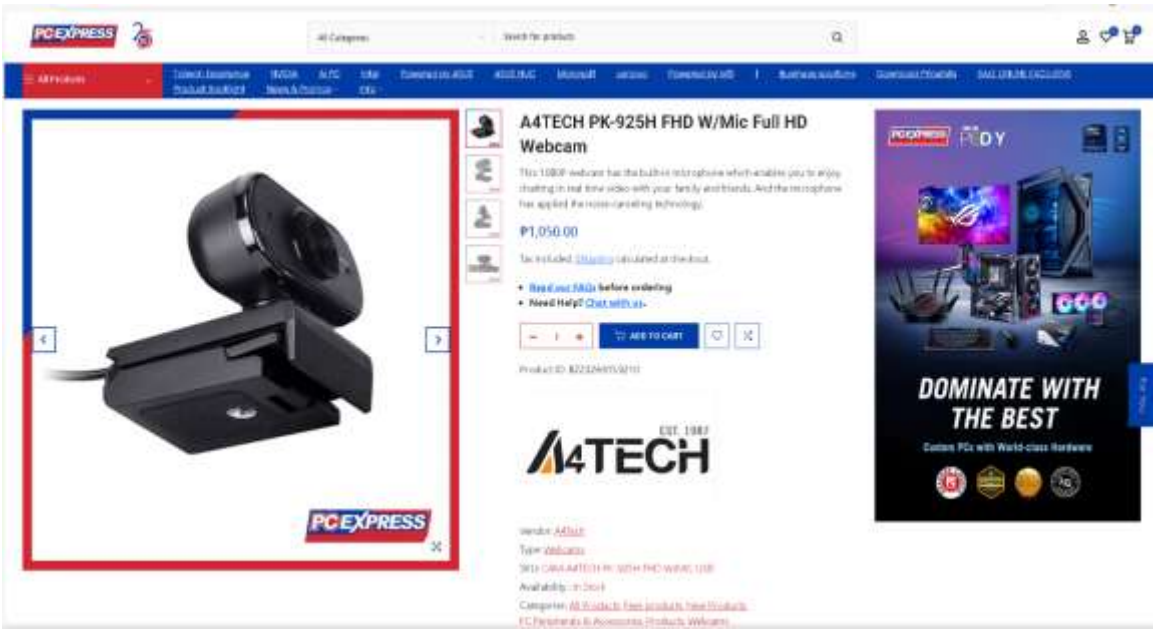
4.5 ★★★★½ | 77 Ratings | 126 Sold | Report

₱50 - ₱113 ~~₱60 - ₱136~~ -17%

Bundle Deals: Any 5 enjoy 2% off

Shipping: Guaranteed to get by 18 Oct >
Free shipping
Get a ₱50 voucher if your order arrives late.

Shopping Guarantee: Free & Easy Returns


PC EXPRESS
A4TECH PK-925H FHD W/Mic Full HD Webcam
₱1,050.00
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Custom PCs with World-class Hardware